

Yogesh Murala

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SUMMARY

AI Engineer with ~3 years of experience building production-grade AI systems and scalable data pipelines. Developed LLM-powered applications, including an autonomous CI/CD debugging agent that reduced manual effort by 60%. Strong in Python, system design, and end-to-end AI integration. Focused on building reliable, high-performance systems for real-world impact.

EXPERIENCE

Senior Engineer - Applied Machine Learning & Generative AI

Bosch Global Software Technologies

August 2023 - Present, Bengaluru

- Architected and deployed production-grade LLM systems using advanced prompt engineering and evaluation strategies, reducing manual triage effort by ~60%.
- Designed and implemented agentic workflows using LangGraph, including stateful orchestration, tool-calling, and failure handling.
- Designed an autonomous CI/CD failure resolution agent integrating LLMs with Azure DevOps and Git pipelines for log analysis, root cause detection, and fix generation using tool-calling and multi-step reasoning.
- Developed multi-step reasoning pipelines leveraging ReAct-style and chain-of-thought prompting to orchestrate log analysis, retrieval, and code-level debugging.
- Implemented modular AI architectures (MCP-style) enabling dynamic tool orchestration across log parsers, vector databases, and knowledge retrieval systems.
- Engineered end-to-end RAG pipelines with intelligent chunking, embeddings, vector indexing (ChromaDB), semantic retrieval, and re-ranking to improve response relevance.
- Built scalable data ingestion and embedding pipelines using Databricks and SQL to maintain consistent and up-to-date knowledge retrieval.
- Developed Python-based APIs for LLM inference, embedding generation, and workflow orchestration with integrated monitoring and validation.
- Orchestrated GenAI workflows using Apache Airflow, including prompt experimentation, evaluation pipelines, and metric-driven validation.
- Designed hybrid AI systems combining deterministic logic and LLM reasoning to improve reliability and controllability.
- Established feedback-driven learning loops to continuously improve model performance and adapt to evolving data.
- Collaborated cross-functionally to deliver scalable AI solutions integrated into developer workflows.

Project Trainee - Machine Learning & Systems Optimization

Bosch Global Software Technologies

September 2022 - July 2023

- Optimized deep learning inference pipelines using Apache TVM (auto-tuning, operator fusion) and NVIDIA TensorRT, improving latency and throughput for real-time AI systems.
- Applied GPU-level optimizations including mixed-precision inference and CUDA acceleration to enhance performance under production constraints.
- Developed and deployed real-time computer vision models using PyTorch and OpenCV, supporting benchmarking and performance analysis.
- Conducted controlled benchmarking and performance analysis of YOLOv8 and the custom multi-head model across Apache TVM and TensorRT, generating actionable insights that informed cross-platform deployment and optimization strategy.
- Partnered with data engineering teams to drive data collection, cleaning, and annotation workflows, ensuring high-quality datasets that improved model accuracy and reliability in production settings.

EDUCATION

B.Tech in Computer Science Engineering (Artificial Intelligence)

Minor in CSE • KL University • Vijayawada, Andhra Pradesh • 2023 • 9.2/10

- Acted as a Peer Mentor where Mentored undergraduate students in Machine Learning, Deep Learning, Natural Language Processing, and Signal Processing.

SKILLS

Programming: Python, C/C++, SQL

Generative AI & ML: LLMs, Transformers, RAG, Prompt Engineering, Fine-tuning (SFT, LORA), NLP

Frameworks: PyTorch, TensorFlow, LangChain, LangGraph, OpenCV

Cloud & Data: Microsoft Azure (Azure DevOps, Databricks), Vector Databases, Embeddings, ETL Pipelines, REST APIs

MLOps: MLflow, Apache Airflow, CI/CD Pipelines, Model Evaluation (Precision, Recall, F1), LLM Evaluation & Benchmarking

Optimization: Apache TVM, TensorRT, CUDA